

EE186 Lab 3: I²C You!

Due: October 22, 2026

1 Introduction

In this lab, you will learn how to interface with an I²C sensor using the LSM303AGR 3-axis accelerometer and magnetometer. I²C (Inter-Integrated Circuit) is a widely used communication protocol that allows multiple devices to communicate over just two wires: SCL (clock) and SDA (data). By the end of this lab, you will:

- Understand how I²C communication works.
- Write an I²C driver to read from and write to the LSM303AGR accelerometer.
- Use the accelerometer to measure gravitational acceleration.

Please find the data sheet for the LSM303AGR [here](#).

Important Note: In this and future labs, you are allowed to use the Hardware Abstraction Layer (HAL) provided by the STM32CubeIDE.

2 Getting Ready

The LSM303AGR sensor is mounted on a small PCB with a voltage regulator and a pin header for easy connection. Before you begin programming, you need to solder the pin header onto the PCB. You can use the soldering stations in Lab 64 or attend office hours if you need assistance. Make sure to double-check your connections before proceeding!

In the rest of this section, you will familiarize yourself with the LSM303AGR datasheet. Carefully study the document to answer the following questions—these details will be essential for writing your I²C driver later in the lab.

Deliverables

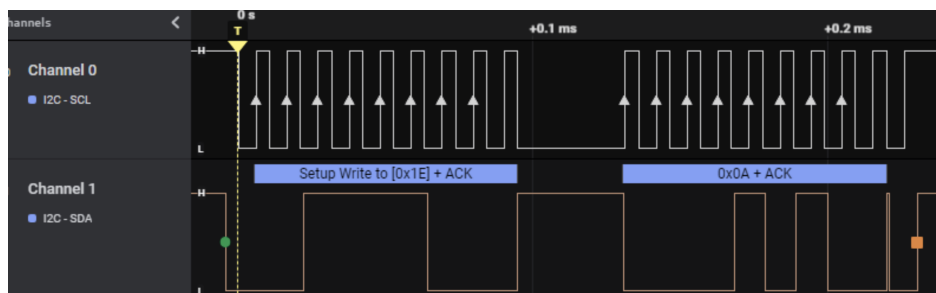
- (1 Point) What is the I²C addresses of the accelerometer and magnetometer? Represent the r/w bit using `b`.
- (1 Point) What is the sub-address (SUB), and how does it differ from the I²C address?
- (1 Point) To enable acceleration data measurement, we need to configure the CTRL1 register by selecting the data rate and setting the enable bits. What is the CTRL1 register address and what value should be written to this register to accomplish this?
- (1 Point) To read acceleration data for the X, Y, and Z axes, what are the 8-bit sub-addresses that the transmitter would use to request the data?

3 Writing the Driver

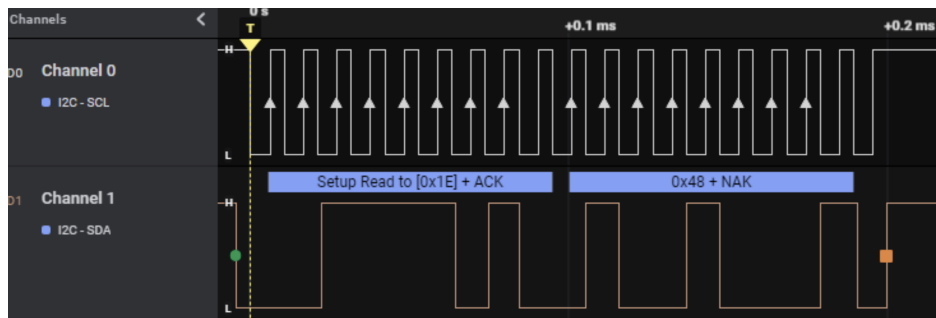
In this section, you will write a driver to read and write from the LSM303AGR. First, place the LSM303AGR on a breadboard and connect its power, GND, SCL, and SDA to the Nucleo board. When connecting power (VCC), use the 3.3V instead of 5V since a higher voltage can damage the sensor.

Whenever you interface with a new sensor, the first step is to verify communication by reading from fixed-value identification registers. These registers contain pre-defined values, allowing you to confirm that your I²C transactions are functioning correctly. Here, you can use some internal device register addresses of the accelerometer by reading values from the `WHO_AM_I` registers as shown in the data sheet Section 8.4 and 8.38.

To read from a register, you must first write the register address to the sensor. This sets up an internal register pointer so that the next read operation retrieves data from the correct location. The I²C transaction consists of one write to the device with its address to establish a connection, then a follow-up write with the address of the internal register you wish to access, in this case, the address of `WHO_AM_I_A` at 0x0F. Here is a trace from the logic analyzer showing the 2 transactions.



Once this is successful, the internal register pointer points to the `WHO_AM_I_A` register. Then, you can read the value of that register with an I²C read transaction. The read transaction starts with an I²C read to the I²C device address, then a follow-up read that will return the contents of the internal device register. Here is a trace from the USB logic analyzer.



In summary, an I²C read is actually composed of two steps: a write to select the register address, followed by a read to fetch the data. In practice, this can be implemented using two separate HAL function calls or with a single combined call that performs both operations under the hood.

Deliverables

- (5 Point) Submit a screenshot of the values read from registers `WHO_AM_I_A` and `WHO_AM_I_M` from the logic analyzer software along with your code.

4 Measuring Gravity

Let's use the LSM303AGR to measure gravity. Your task is to read the X, Y, and Z acceleration values and display them on a terminal.

Here are some things to consider. Make sure to enable acceleration measurements and start simple by reading one axis at a time. Read raw values to start. The value is a 16-bit number and you can read them in one I²C read by adjusting the size argument. The 16-bit acceleration value is a 2's complement number. The accelerometer will provide a value from -1G to 1G depending on the axis orientation to gravity. That is, when the axis is oriented parallel to gravity, it will produce either -1G or 1G. When the axis is perpendicular to gravity, it will produce zero.

Deliverables

- (5 Point) By changing the orientation of your sensor, show that you can measure gravity from -1 to 1 G in all three axes. The output should be displayed on a terminal like the Arduino sketch terminal. Display both the raw values and converted values in Gs for each axis. Submit a video showing both sensor movement and terminal output, along with the code.